

Airframe Technology Master

Subject 05: CATIA-DMU-DIGITAL

Lecture 21

DOCUMENTATION

FULL 3D Methodology & Key Processes

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3 July 2025

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Glossary:

A/C	Aircraft
CSG	Catia Single Gate
DMU:	Digital Mock-Up
FDQ	Full Design Quality
FTA:	Functional Tolerancing & Annotation
MSN	Manufacturing Serial Number
PMI	Product Manufacturing Information

Introduction to the lecture



On the Lecture 01, we do key remarks for this subject and introduce:

- Documentation systems & their differences: 3D as master & FULL3D
- Digital Mock-Up: Global environment where develop all processes of product.
- FULL3D concept as the previous KEY step to develop the Digitalization.

After, we showed on Lectures 02 to 20, the modelling methods in some disciplines (machined, sheetmetal, composites, assemblies) by the use of CATIA.



On next Lectures 21 to 26, we will give a simple, practical & real application of these methodologies on the program & specially dedicated to this Airframe master:



- How is applied the generation & documentation processes.
- How the information is introduced & checked.
- How is applied the release process for validation of this information.

All of them represents a BIG “change mindset” & “a new age of this business”.

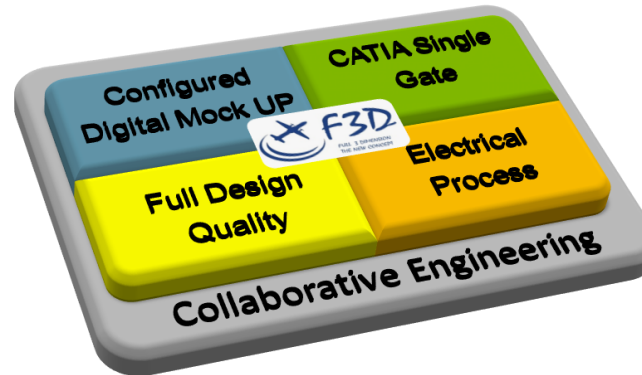
Introduction to FULL3D Methodology



Full 3D

FULL 3D overview

- FULL 3D is much more than remove Drawings, is a work methodology that allows to define the Digital Mock-Up of product completely integrated & available for ALL
- Much more than format change,... it is “Cultural Change” and a MINDSET



FULL 3D Key processes

- ❖ Create & provide a work environment
- ❖ Define & Include the information
- ❖ Control & Validate the information
- ❖ Universalize the Concept
- ❖ Extend the “Concept for All”

CDMU - Configured Digital Mock-Up
FULL3D & CATIA Single Gate (CSG)
FULL Design Quality “Knowledge”
FULL3D & Electrical Process
FULL3D & Collaborative Engineering

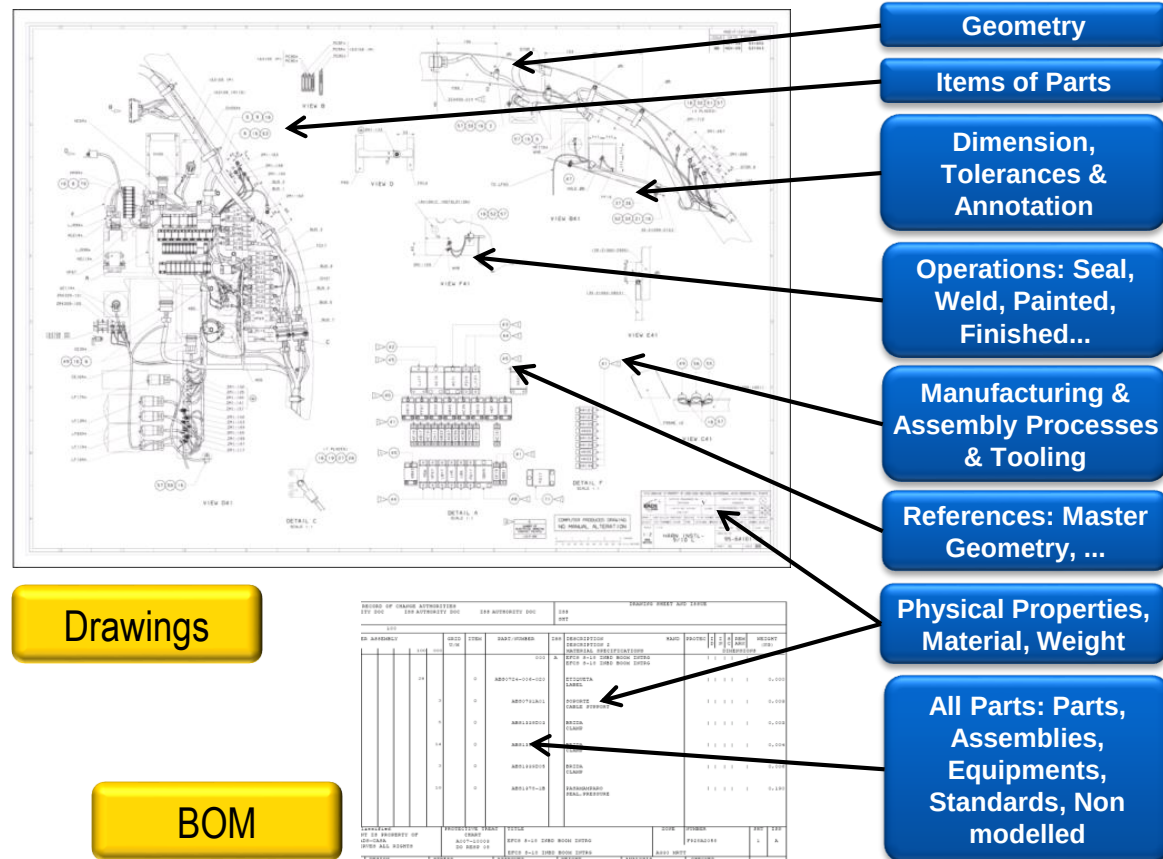


What means FULL3D..?

- ❖ Generate a Documentation system from the **Digital Mock-Up** including in 3D all the information that define the Product, making it unnecessary the drawing.
- ❖ Generate an automatic representation in a **Light Viewer** with the “solid enriched with information” that can be consult & use from “any place” (area & function).
- ❖ Simplification (**LEAN concept**) of the information that in the past was included in the drawing and other “places”, normalizing the processes & methods.
- ❖ **Master-Slave concept**: Data is included in "only on a site" and “auto & robust” replicated in other sites & tools.

Data from Content point of view

Traditional drawing & BOM - Only as a reference of the content



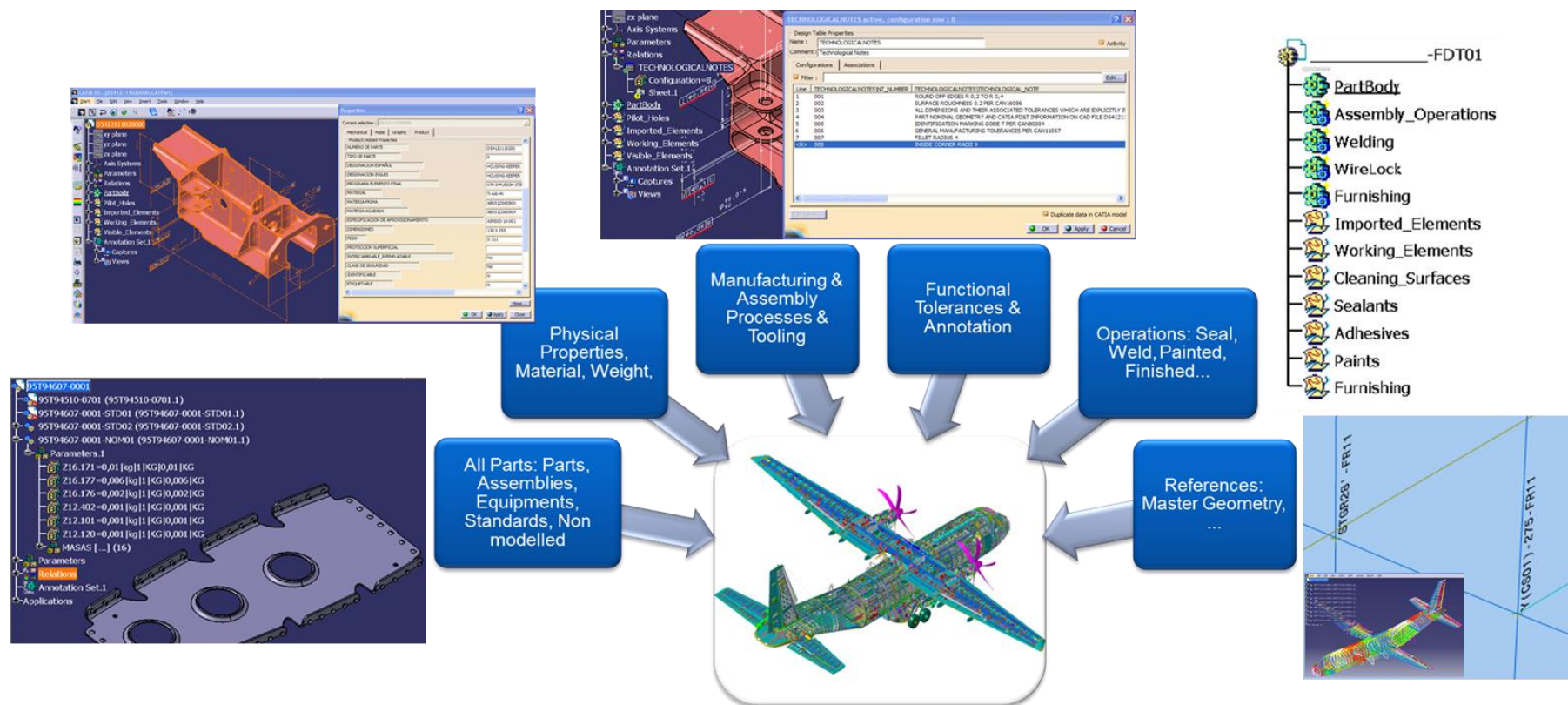


MCH-D5

What is introduced in FULL3D..?

All information necessary to define, certify, industrialize, test, validate & maintain the Product

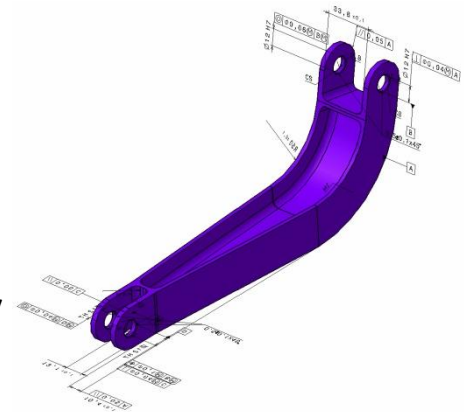
- All Parts & PMI Information.
- Finishing & Assembly Operations



All Parts

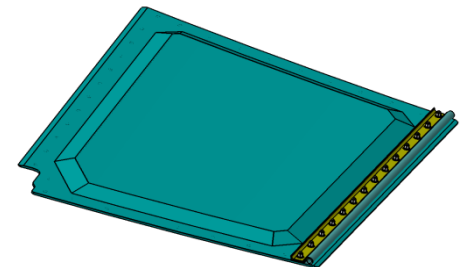
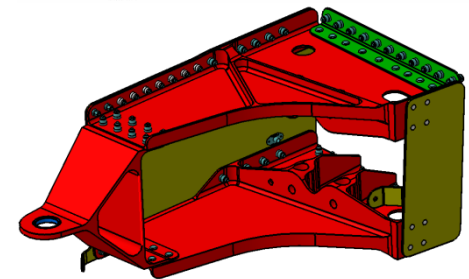
Identifiable parts

- All part identified with the Part Naming criteria of the A/C.
- An identifiable part can be represented by a unique solid or group of solids, standards parts, etc.
- For its complete definition or production, could be necessary other files: skins, surfaces, ply contour, etc.



Special notes & examples:

- Envelop part as result of composition of some materials (composite part as result of fibers & resins).
- Finished part as result to machine a Raw part (casting, forging, etc.)
- Reworked part as result to machined an original part.
- Flexible part as result of different disposition/shape, etc.
-



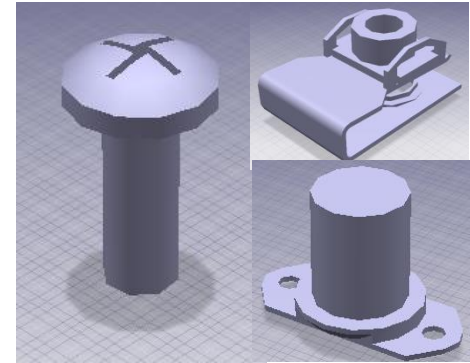


AH&F

All Parts (cont.)

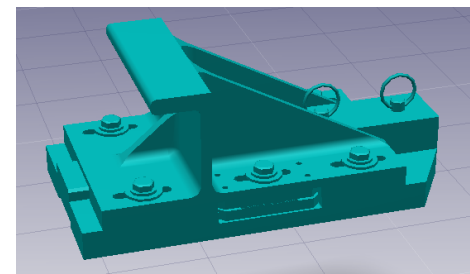
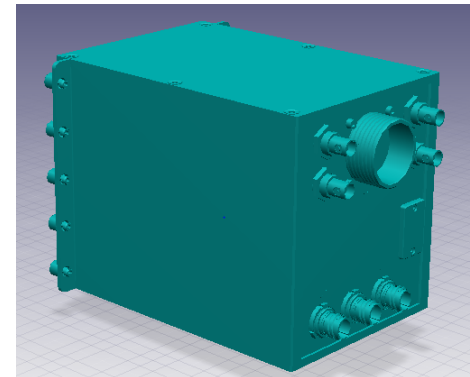
Standard part

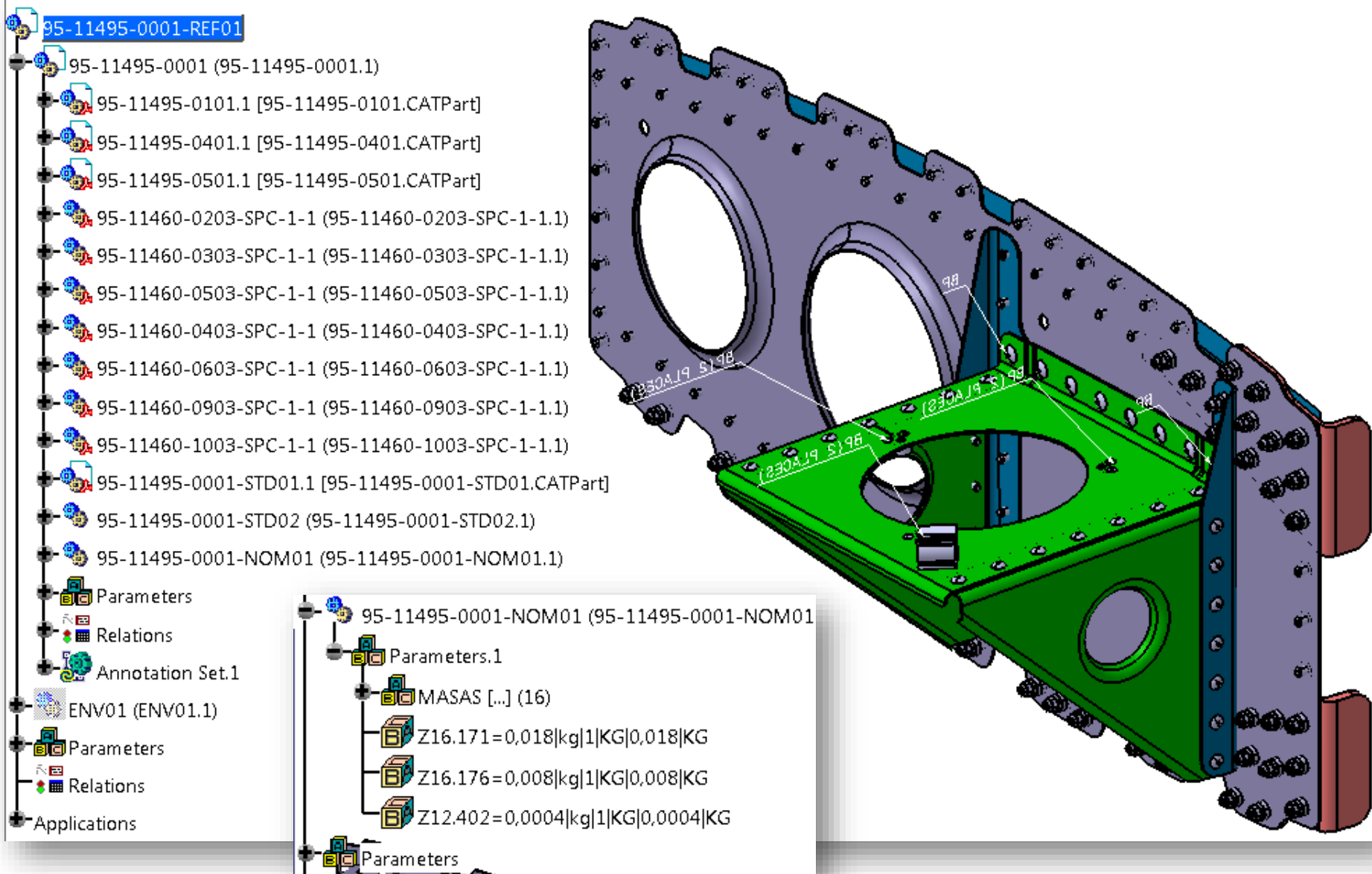
- Standardize part like: bolt, rivet, sleeve, connector, etc.
- Its geometry is available & loaded from a catalogue of A/C.
- Whole Part Number identify an unique part (all properties).
- Note: Enriched content for usability (rivet)
- Risk: it can be used the same P/N with different criteria.



Equipment

- Non-standard part provides by a supplier & installed in A/C.
- Geometry includes whole equipment definition as assembly.
- Available & loaded from a catalogue or database.
- It includes the external geometry (at level of detail) & all the interfaces for its installation (functional & structural).
- Reference at Standard interfaces & functionality.





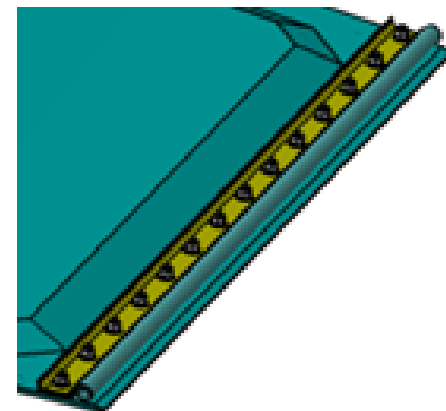


NOM-95T

All Parts (cont.)

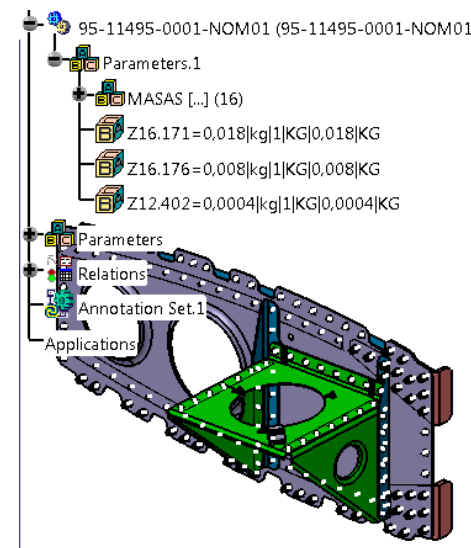
Semi-finished part

- Part defined made from semi-finished part: tape, rubber, etc.
- Usually it has geometry named from a standard following with a code that represent its length or other geometrical characteristic.



Non-Modelled Part

- Part without geometry & included in assembly: adhesive, sealant, etc.
- Generally called consumable, it is identified by a standard plus a quantity with reference an unit (length, weight, etc.)
- Usually it has a related geometry of place where it is applied.
- Special element taking into account in the assembly process and consideration in Full3D.





PMI-95-1

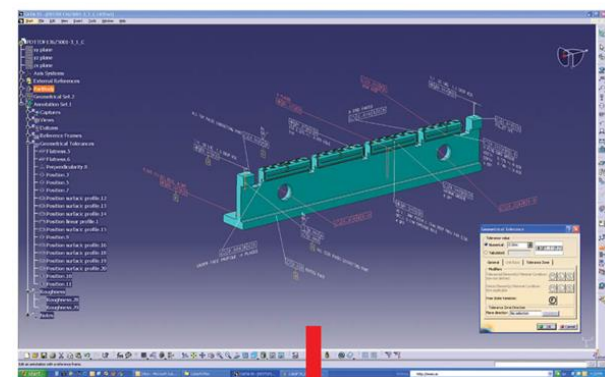
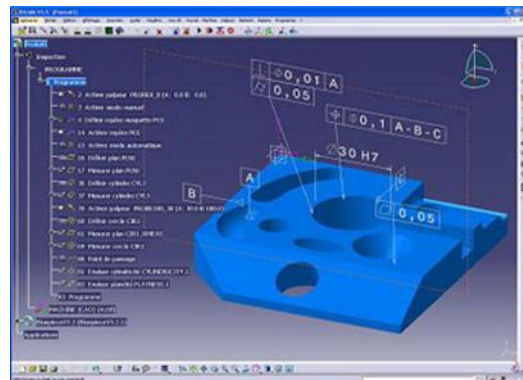
PMI - Product Manufacturing Information

Definition

- PMI represents the information that is necessary for manufacture & assemble the parts, & traditionally were included in Drawings & in Bill Of Material.
- PMI is a key topic in the FULL3D information (collaborative engineering)

Include:

- ❖ Physical properties & Material specifications.
- ❖ Technological Notes that define manufacturing & assembly processes.
- ❖ Geometric dimensions & tolerances, 3D annotation (text), etc.
- ❖ Surface & finished operations.



PMI - Physical properties and Material specifications

Definition

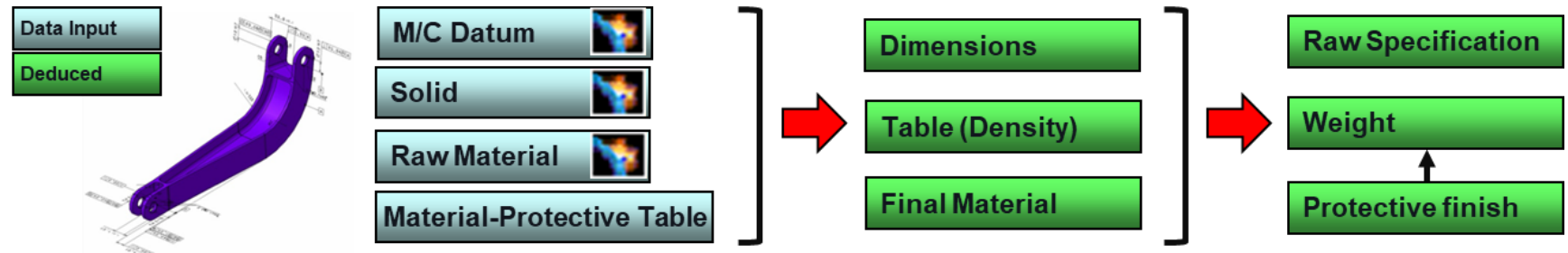
- This chapter represent a heterogeneous group of information about of Physical properties and Material specification of the part.

Include:

- ❖ Material definition: Raw material, Final material, etc.
- ❖ Material specifications: Dimensions, raw specifications
- ❖ Protective finished: Finish treatment, etc.



KEY point: to establish the relation between all data to obtain them by deduction (part of Full Design Quality, explain in lecture 22)





CSG-95T

CATIA Single Gate (CSG) - Metadata

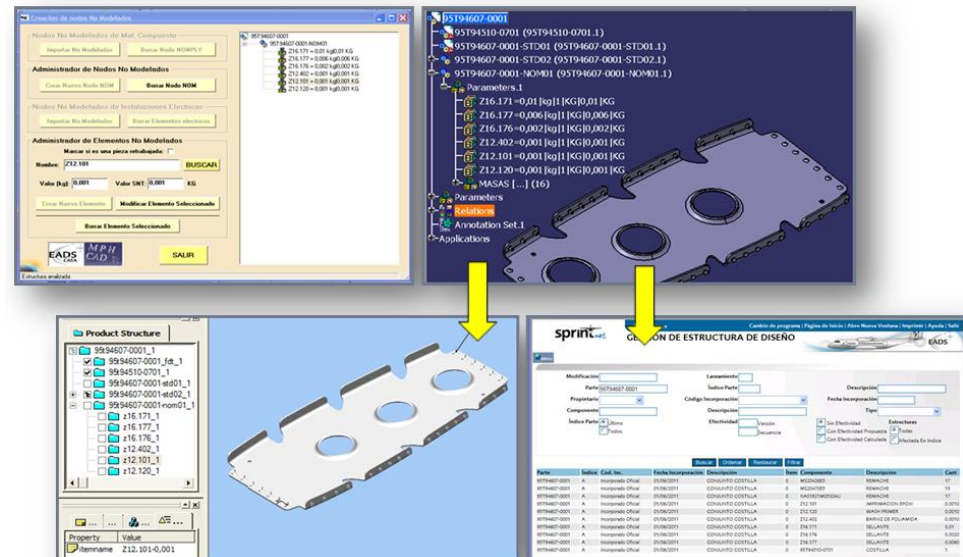
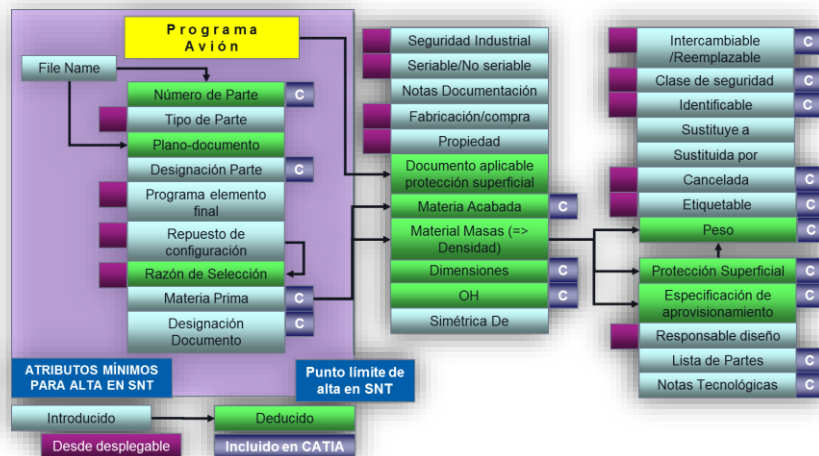
There is an information called metadata, that generally was included in the drawing title block or the Bill Of Material, and that represents relations between parts, remark to the operation, data from its supply, etc.



- All data are included in “Only a site - CATIA” (geometry & metadata).
- Unique & Direct access to single database (material, protections, etc.)
- Master-Slave concept: “Auto & Robust” replication in other tools



What data but take part of PLM toolset & what data must be included in CATIA?





CSG-95T
DD-D5

Catia



CSG

“Master-Slave concept”

PLM

sprint.net

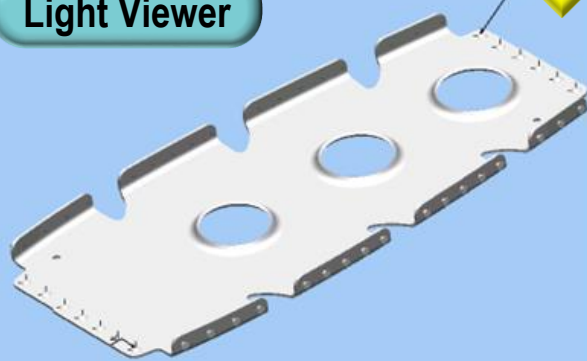
Acciones: Cambio de programa | Parámetros | Inicio | Abre Nueva Ventana | Imprimir | Ayuda | Salir

GESTIÓN DE ESTRUCTURA DE DISEÑO

Modificación: Lanzamiento: ☐
 Parte: 95T94607-0001 Índice Parte:
 Propietario: Código Incorporación: Descripción:
 Componente: Descripción: Fecha Incorporación:
 Índice Parte: ☐ Último ☐ Todos Efectividad: ☐ Versión ☐ Secuencia Tipo:
☐ Sin Efectividad ☐ Con Efectividad Propuesta ☐ Con Efectividad Calculada Estructuras: ☐ Todas ☐ Afectada En Indice

Parte	Índice	Cod. Inc.	Fecha Incorporación	Descripción	Item	Componente	Descripción	Cant.
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	M520426E5	REMACHE	17
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	M520470E5	REMACHE	13
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	NAS1921M05504U	REMACHE	17
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z12.101	IMPRESION EPOXI	0.0010
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z12.120	WASH PRIMER	0.0010
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z12.402	BARVIZ DE POLIAMIDA	0.0010
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z16.171	SELLANTE	0.01
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z16.176	SELLANTE	0.0020
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	Z16.177	SELLANTE	0.0060
95T94607-0001	A	Incorporado Oficial	01/06/2011	COIUNTO COSTILLA	0	95T94510-0701	COSTILLA	1

Light Viewer



Creacion de nodos No Modelados

Importar No Modeladas Borrar Nodo NOMPLY

Administrador de Nodos No Modelados

Crear Nuevo Nodo NOM Borrar Nodo NOM

Nodos No Modelados de Instalaciones Electricas

Importar No Modeladas Borrar Elementos electricos

Administrador de Elementos No Modelados

Marcar si es una pieza retrabajada: ☐

Nombre: Z12.101 BUSCAR

Valor (kg): 0.001 Valor SNT: 0.001 KG

Crear Nuevo Elemento Modificar Elemento Seleccionado

Borrar Elemento Seleccionado

EADS CASA MPH CAD

SALIR



NoT-D5

PMI - Technological Notes

Information related with Processes (what & how) & Standards (free tolerances, etc.) that must be performed & compliance during the Manufacturing, Assembling & Testing of the Part.

Harmonization
Simplification
Standardization
Parametrization
Automation



In FULL3D, they have the following characteristics:

- Harmonized, normalized & included in a specific standard.
- Organized by processes & disciplines (machined, sheetmetal, composite, etc.).
- They can not be edited except specific & controlled “fields” in its description.
- Applied by the user in the assistant way, applied by group with: obligatory, optional & complementary.
- Related with other data including in the 3D model.
- Content & geometry control explained in lecture 22

CODE	TEXT / DESCRIPTION
IDXX-YY	FILLET RADIUS ##
IDXX-YY	INSIDE CORNER RADII ##
IDXX-YY	SHOT PEENING PER I+D-P-XXX
IDXX-YY	GRAIN DIRECTION OPTIONAL
IDXX-YY	COLOUR GREY ##
IDXX-YY	* LOCATION OF RIVETS FOR ANCHOR NUTS
IDXX-YY	+ PILOT HOLE DIA 2.5
IDXX-YY	FOR EXACT GRAIN DIRECTION SEE M/C DATUM AXIS PER CAN10605
IDXX-YY	SPECIFIC SURFACE ROUGHNESS ## PER CAN160XX INCLUDED IN 3D MODEL
IDXX-YY	CENTRE HOLE DIA ## PER DIN332 TYPE ##
IDXX-YY	SHARP EDGES AND EXEMPT OF BURRS
IDXX-YY	CHAMFER AT ASSEMBLY
IDXX-YY	SHOT PEENING PER I+D-P-XXX, INTENSITY ##
IDXX-YY	ELECTRICAL BONDING SURFACE INCLUDED IN 3D MODEL
IDXX-YY	THEORETICAL DATA PER DRAWING/MODEL ##
IDXX-YY	MASK BEFORE PAINTING WITH ## EXCEPT IN MARKED ZONE
IDXX-YY	LAYER THICKNESS ##
IDXX-YY	DIMENSION PRIOR TO PROTECTIVE FINISH CODE ##: ## MIN - ## MAX
IDXX-YY	LINEAR TOLERANCES ##
IDXX-YY	THICKNESS TOLERANCES ##
IDXX-YY	THICKNESS TOLERANCES +/- 0.15
IDXX-YY	LINEAR TOLERANCES +/- 0.2



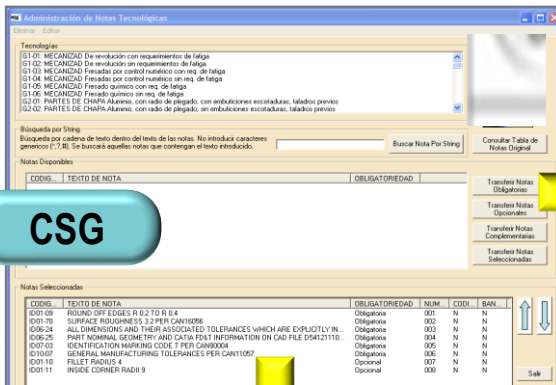
DD-D5

PMI - Technological notes (way of working)

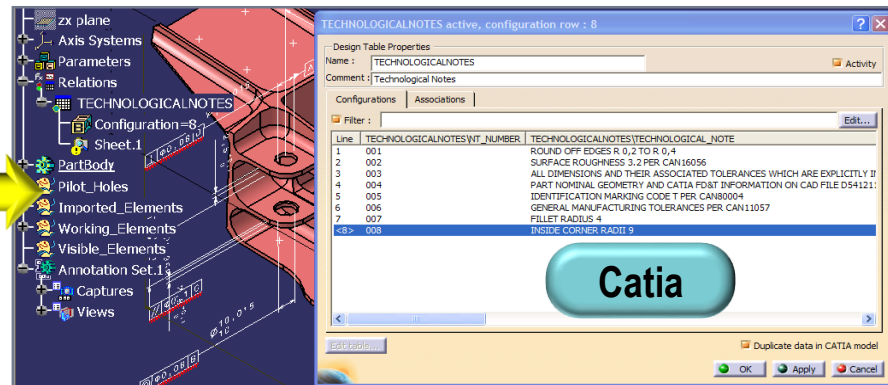
Utility to assist way
Rules & Database Link

Inclusion in 3D model (Catia)
Control of the content & link

Harmonization
Simplification
Standardization
Parametrization
Automation



CSG



Catia

“Master-Slave concept”

Programa: K Usuario: RUIZ SANCHEZ, MIGUEL P. Factoria: - N4 OTRAS NOTAS - Windows Internet Explorer

Inicio | Página de Inicio | Abre Nueva Vent

Tipo	Código Nota	Texto	Codificado	Cancelado
G	001	ROUND OFF EDGES R 0,2 TO R 0,4	☐	☐
G	002	SURFACE ROUGHNESS 3.2 PER CAN16056	☐	☐
G	003	ALL DIMENSIONS AND THEIR ASSOCIATED TOLERANCES WHICH ARE EXPLICITLY INDICATED	☐	☐
G	004	PART NOMINAL GEOMETRY AND CATIA PDM INFORMATION ON CAD FILE D5412111020000	☐	☐
G	005	IDENTIFICATION MARKING CODE T PER CAN80004	☐	☐
G	006	GENERAL MANUFACTURING TOLERANCES PER CAN11057	☐	☐
G	007	FILLET RADIUS 4	☐	☐
G	008	INSIDE CORNER RADII 9	☐	☐

PLM

RECORD OF CHANGE AUTHORITY

MOD NO	QUANTITY	REQ	APPROV	QD	ITM	PART NUMBER	DESCRIPTION	NAME	PROTEC	DATE	REVISION
210						210	DESCRIPTION 2				
							MATERIAL IDENTIFICATION				
							DESCRIPTION 3				
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PLM

Export in Automatic way
to Config/PLM database

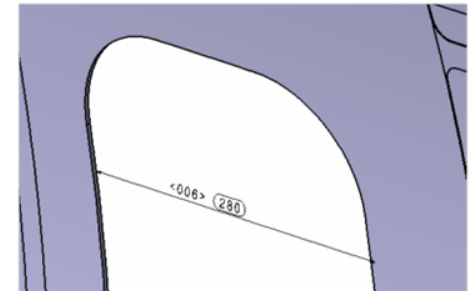
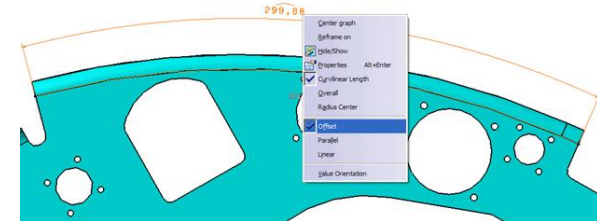
Output to Bill Of Material of
the Part (if was necessary)



PMI - Functional Tolerancing & Annotation (I) – Dimensions y Tolerances FTA-D5

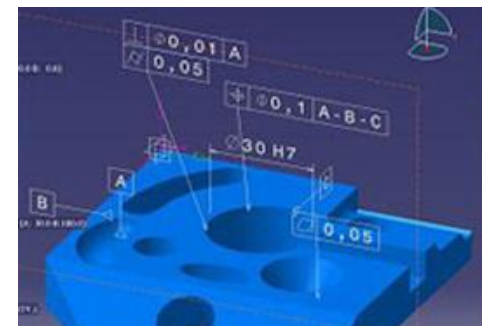
Dimensions:

- General dimensions are not included except specific cases (non-typical): length, angles, etc. (i.e. flange in SheetMetal)
- Other dimensions included: Interface points & dimensions for interchangeability.



Tolerances:

- General tolerances are not included because are indicated by reference to standard.
- Only specific tolerances are included.
- Geometrical Dimensioning & Tolerances GD&T



There is a Standard that define what, how & where to define this properties (FTA), base for Parametrization & Automation

Harmonization
Simplification
Standardization
Parametrization
Automation



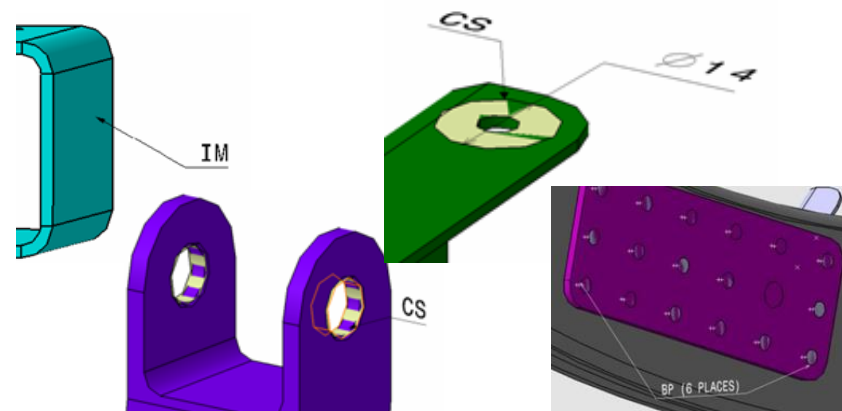
PMI - Functional Tolerancing & Annotation (II) – Zones & References

FTA-95-11

SHE-95T

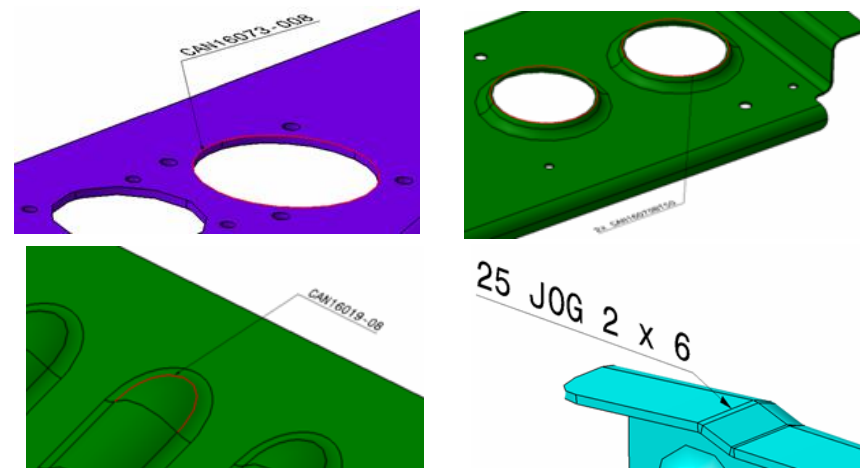
Identification for zones & specific operations

- Identification Marking (IM)
- Cleaning Surface (CS)
- Bonding point (BP)
- Tooling holes (TH)



Reference to standard operation

- Lightening holes
- Cut-outs & Swages
- Joggles.
- Thread representation



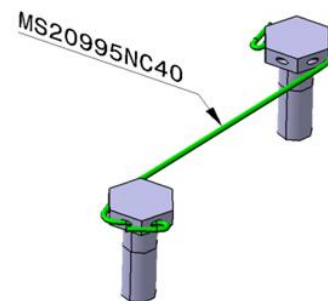


FTA-95-11

PMI - Functional Tolerancing & Annotation (III) – Text with leader

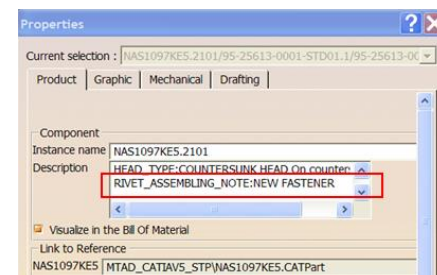
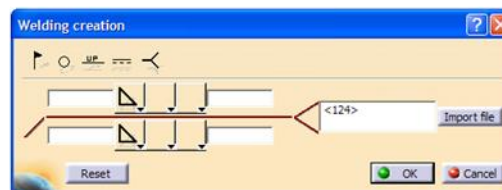
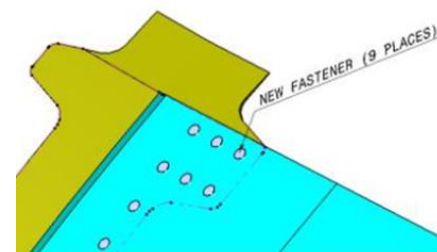
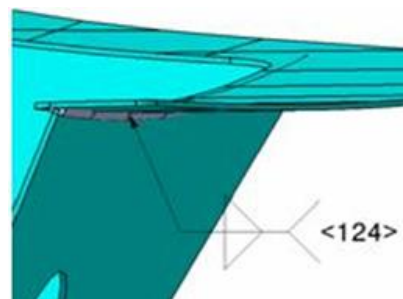
Texts as “Flags” to express reference to parts or operations

- Wire lock installation and tape furnishing
- Sealant & Adhesives
- Paints & ...
- Lubricants & ...



Text as reference plus criteria (Flags)

- Previous Holes
- Rivets installations
- Welding



Finishing and Assembly Operations (I) - Definition

Definition



- Define the operations that are performed to or over the part generally when the assembly is mounted or is installed in the A/C.
- Includes the information of operation & their geometry associated.
- This geometry does not belong to the part at definition level (in storage mode).



- Each installation can have several assembly operations in its definition.
- All geometry & information are included at “CATPart file” with the reference of the installation.
- In Installation: Operation Responsibility vs Part Number Responsibility



- These operations could be applied at same assembly (Part Number) but at different position and at different effectivity.
- Note: Instance & effectivity concept, explained later (Data Quality lecture)

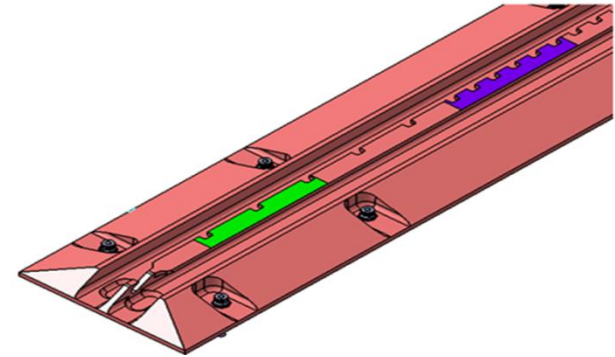


DRL-CUT

Finishing and Assembly Operations (II) - Examples

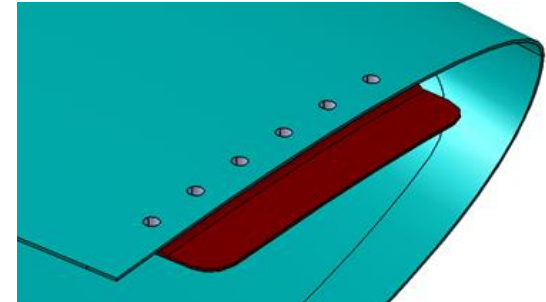
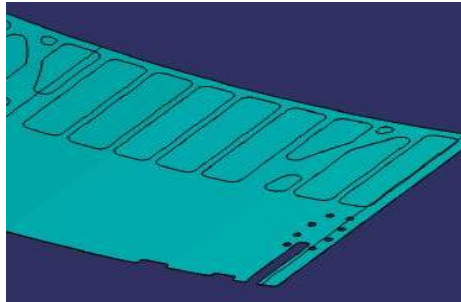
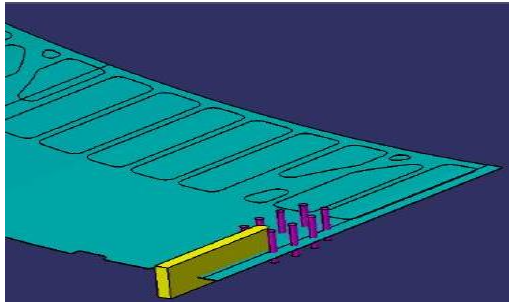
Finishing operation

- Cleaning: Surface that is clean for later operation
- Sealing: Edge or surface where the seal is applied
- Painting: Surface where the paint is applied
- Welding: Edge or point where the welded is applied
- Finished: Surface with specific protective finish.



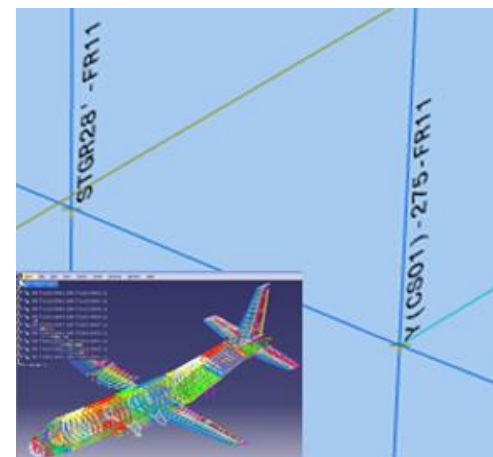
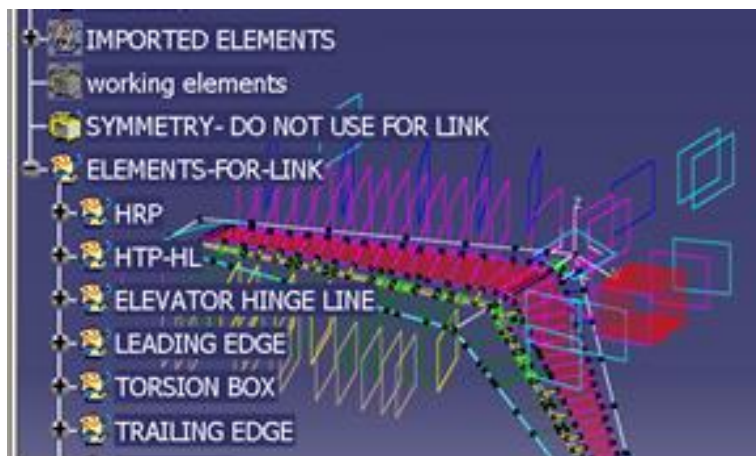
Assembly Operations

- Cutting: Geometry that represent the machined operation
- Drilling: Geometry that represent the drilling (holes) operation



External References

- Include the information necessary for location & reference of part on the A/C.
- Even though in the DMU are included all parts and they can be seen in Context; when a part have to be document in isolation way, usually it is necessary to include references about situation of this part in its installation.
- The common way is to include theoretical lines of Master Geometry belong to the Structural Diagram as reference (frames, stringer, etc.).
- All geometry is included at “CATPart file” with the reference of the installation.

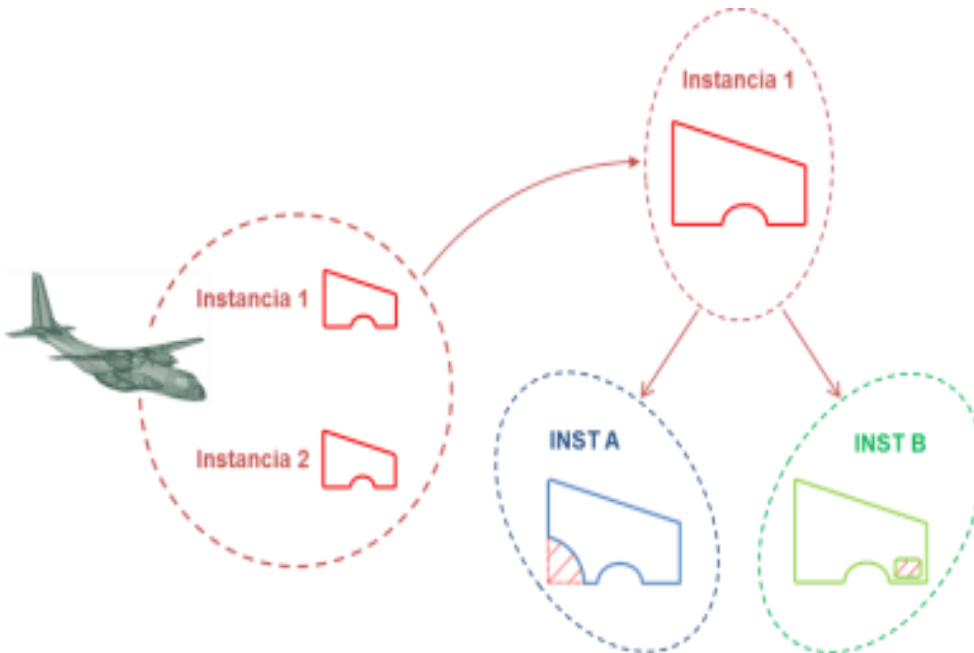


Installation: CATProduct REF case, explained later (Release Process lecture)

Instance Part Concept in FULL3D

Instance Part concept – Definition & rational

- Instance is the specific representation of one part in a specific position in a A/C.
- The Instances of a Part share the same P/N, but they can have applied different operations and/or properties in the A/C, what distinguishes them from the rest.
- General data that define a Instance are: P/N, geometry, position & effectivity.





DRL-CUT

Instance Part Concept in FULL3D (cont.)



Examples to distinguish:

- A part can have different protection finish & *assembly operation* according to its position & effectivity.
- An standard label can have different Text according its function in A/C.
- A rivet can have different assembly condition according to its position in A/C.
- The way to distinguish them is using the assembly properties of Instance.



This “Distribution of Instances”

- It must be taking into account in the A/C configuration.
- The workflow for approval of parts must be performed with this consideration.
- The Digital Product Continuity need to assure the industrialization & support are performed according to the particularity and properties of each Instance.



It represent “the new concept” to manage the A/C configuration & Way of Working.

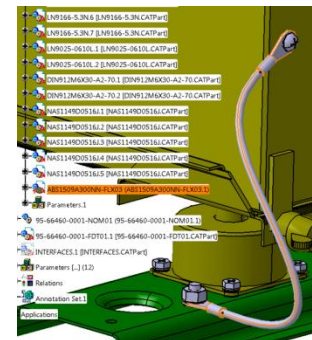
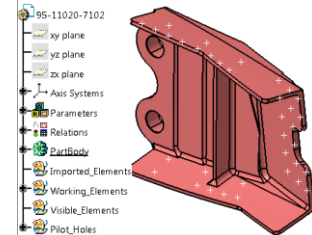
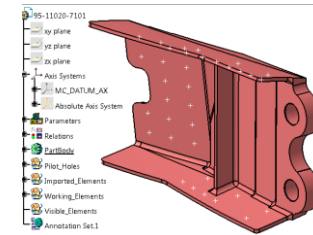


FTA-95-11

Specials representation of parts in FULL3D

Specific representation of Parts:

- Symmetrical parts complete modelled but documentation limited.
- Flexible parts sheetmetal, springs, etc.
- Deformable parts foam, etc.
- Semi products geometry and quantity



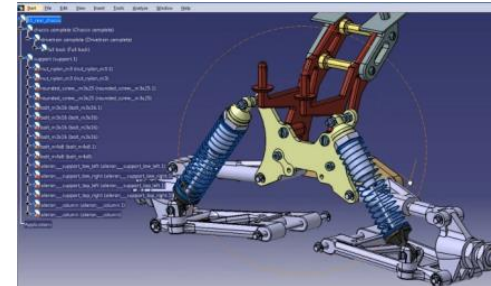
Parts Made From:

- Reworked
- ...

? Catia Tree vs Product Structure vs BOM

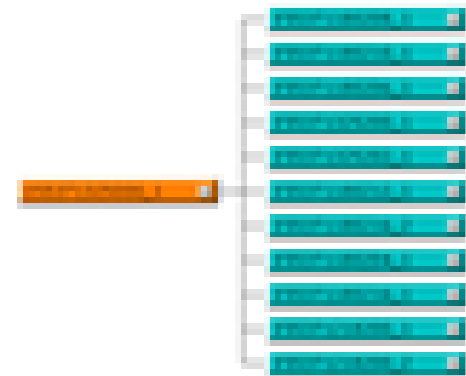
CATIA Tree

- Hierarchical breakdown of a product expressed by Catia elements (files/data) used to define/represent the product
- The Data disposition is mainly according to the CATIA module (machined, composite, etc.) of the SW provider.



Product Structure

- Hierarchical breakdown of a product in term of parts, used to represent the product in a system or database.
- It establish the Parent/Children link for documentation, configuration and change management.



BOM (Bill of Material)

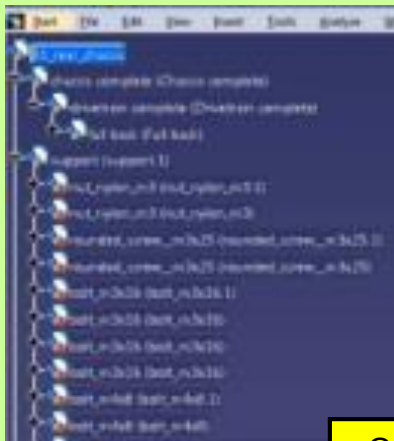
- *Specific list* of parts, materials, components, standards and quantities of them, needed to manufacture the Part.
- Joint with them, are included other “metadata” of Part.

BOM OF CHASSIS AUTOMATIQUE									
REV. 001				REV. AUTOMATIQUE 001				REV. 001	
NO. RANGEMENT	LIBELLE	QTE	UNITE	NO. RANGEMENT	LIBELLE	QTE	UNITE	NO. RANGEMENT	LIBELLE
1	CHASSIS AUTOMATIQUE	1	PIECE	1	CHASSIS AUTOMATIQUE	1	PIECE	1	CHASSIS AUTOMATIQUE
2	BOITIER	1	PIECE	2	BOITIER	1	PIECE	2	BOITIER
3	BOITIER	1	PIECE	3	BOITIER	1	PIECE	3	BOITIER
4	BOITIER	1	PIECE	4	BOITIER	1	PIECE	4	BOITIER
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Catia Tree vs Product Structure vs BOM (Cont.)

CATIA Tree

- Catia Part file level

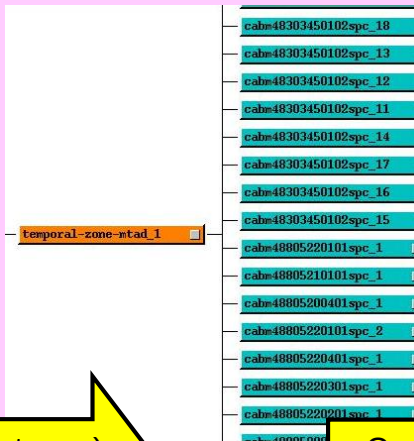


• Conversion (File/Instance)
• Trazability & Configuration

- Geometrical segregation for representation.
- Extended files use.
- All parts defined (F3D)

Product Structure

- Instance Part level



• Conversion (Instance/P/N)
• Trazability & Configuration

- Segregation at Instance Part level: P/N + Position
- All parts defined (PLM)
- Properties for Instances.

BOM (Bill of Material)

- P/N level definition

ITEM	PART NUMBER	QTY	DESCRIPTION
1	ABR0124-014-020	1	STANDARD LARRE
2	ABR0124-014-020	1	STANDARD LARRE
3	ABR0124-014-020	1	STANDARD LARRE
4	ABR0124-014-020	1	STANDARD LARRE
5	ABR0124-014-020	1	STANDARD LARRE
6	ABR0124-014-020	1	STANDARD LARRE
7	ABR0124-014-020	1	STANDARD LARRE
8	ABR0124-014-020	1	STANDARD LARRE
9	ABR0124-014-020	1	STANDARD LARRE
10	ABR0124-014-020	1	STANDARD LARRE

- Segregation at P/N level
- Quantity per P/N
- P/N <=> Metadata
- Documental properties.



TOOL

DATABASE



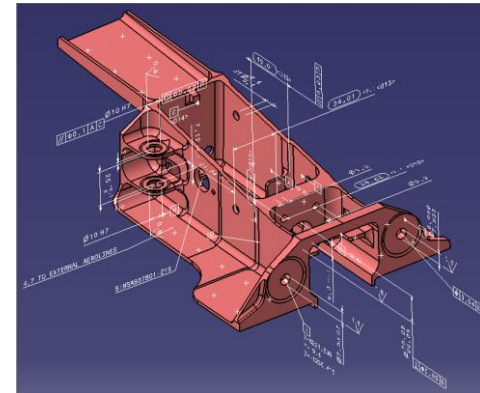
MCH

SHE

Data of Element/Component – Elementary part

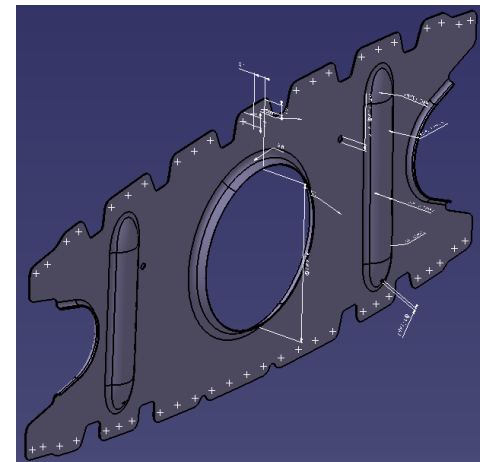
Machined part

- Single & unique part
- Generate by Part Design module
- Include all shapes that for CNC – Numerical Control.
- Specific methodology for feature disposition in the solid tree
- Feature groups for typologies (raw part, pockets, cutting, holes, etc.) to facilitate the part machining.



Sheet Metal part

- Single and unique part
- Generate by Sheetmetal Aerospace module
- Include all folder & unfolded part
- Specific technological features: joggles, lightening, etc.
- Technological tables with properties for shaped

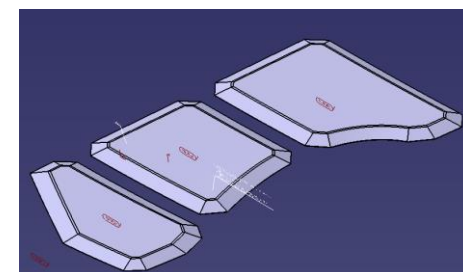
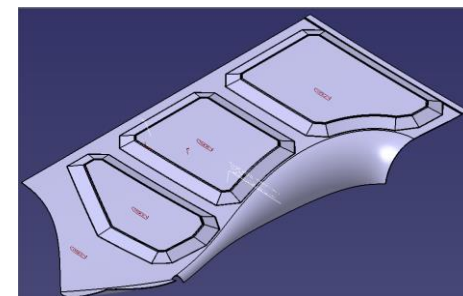
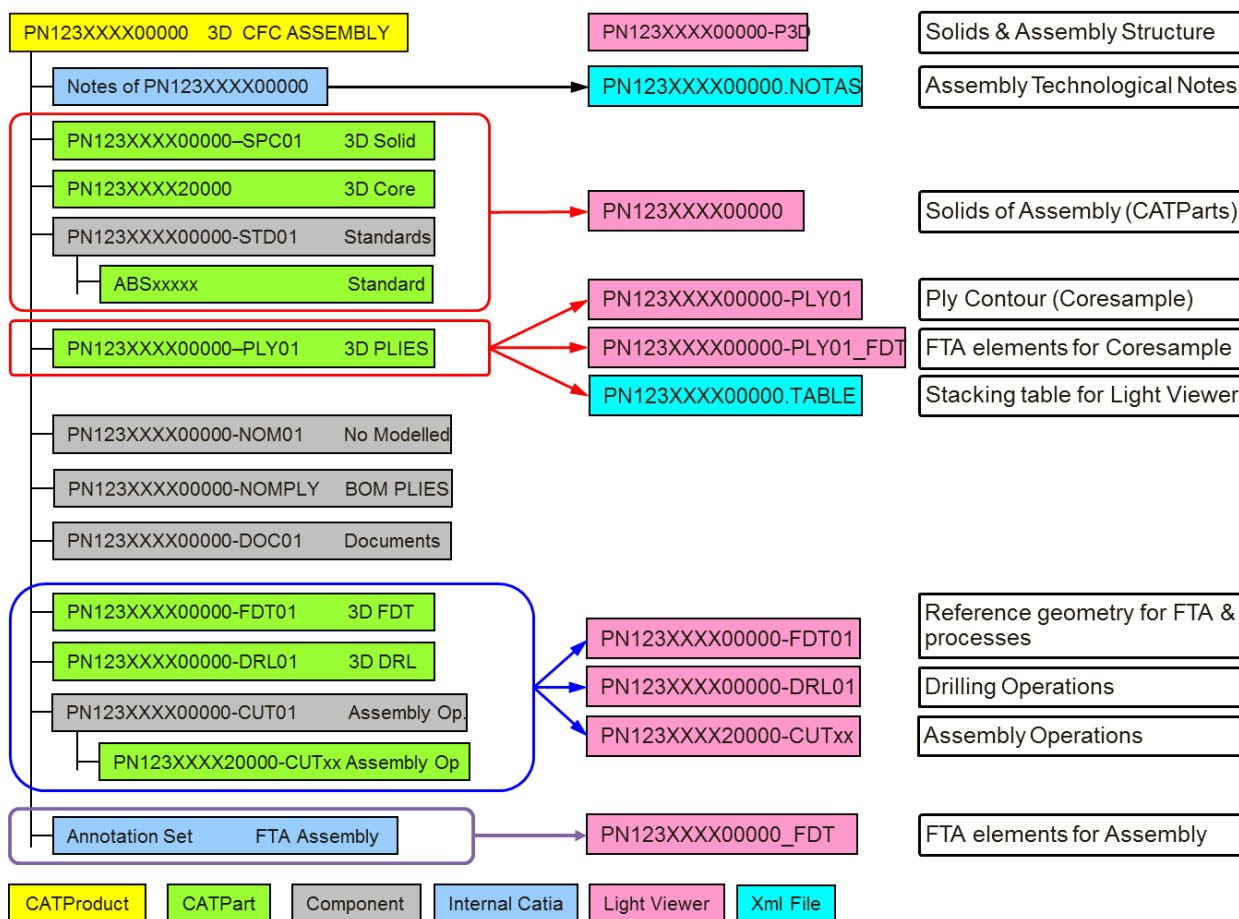




CPD

Data of Element/Component – CFC Assembly part

Composite part: Single & unique part composed by “the union of materials”

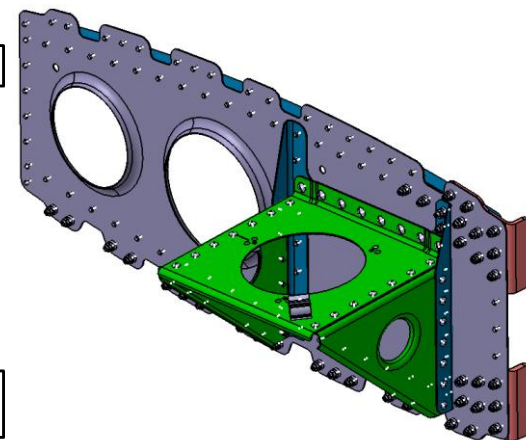
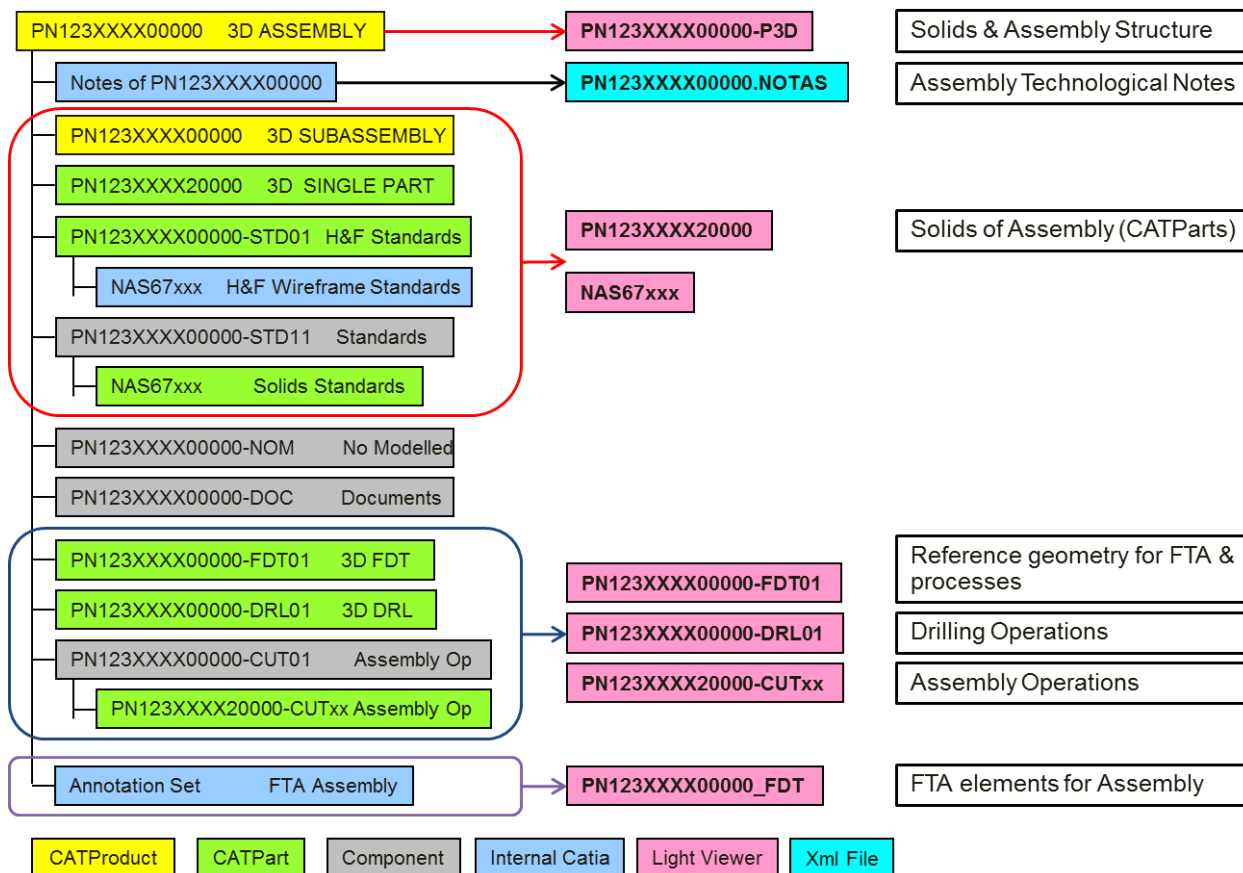




ASSY

Data of Element/Component – Assembly part

Assembly: Usually it represents an sub-assembly storable part

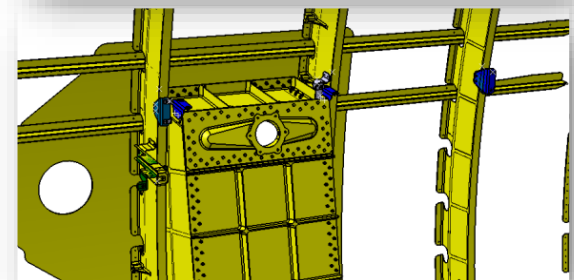
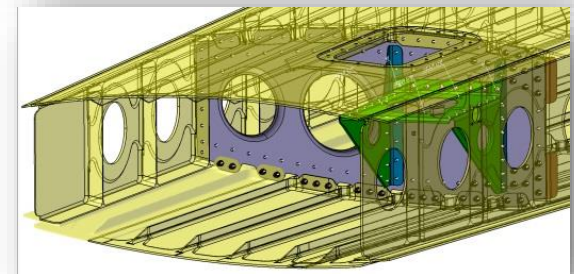
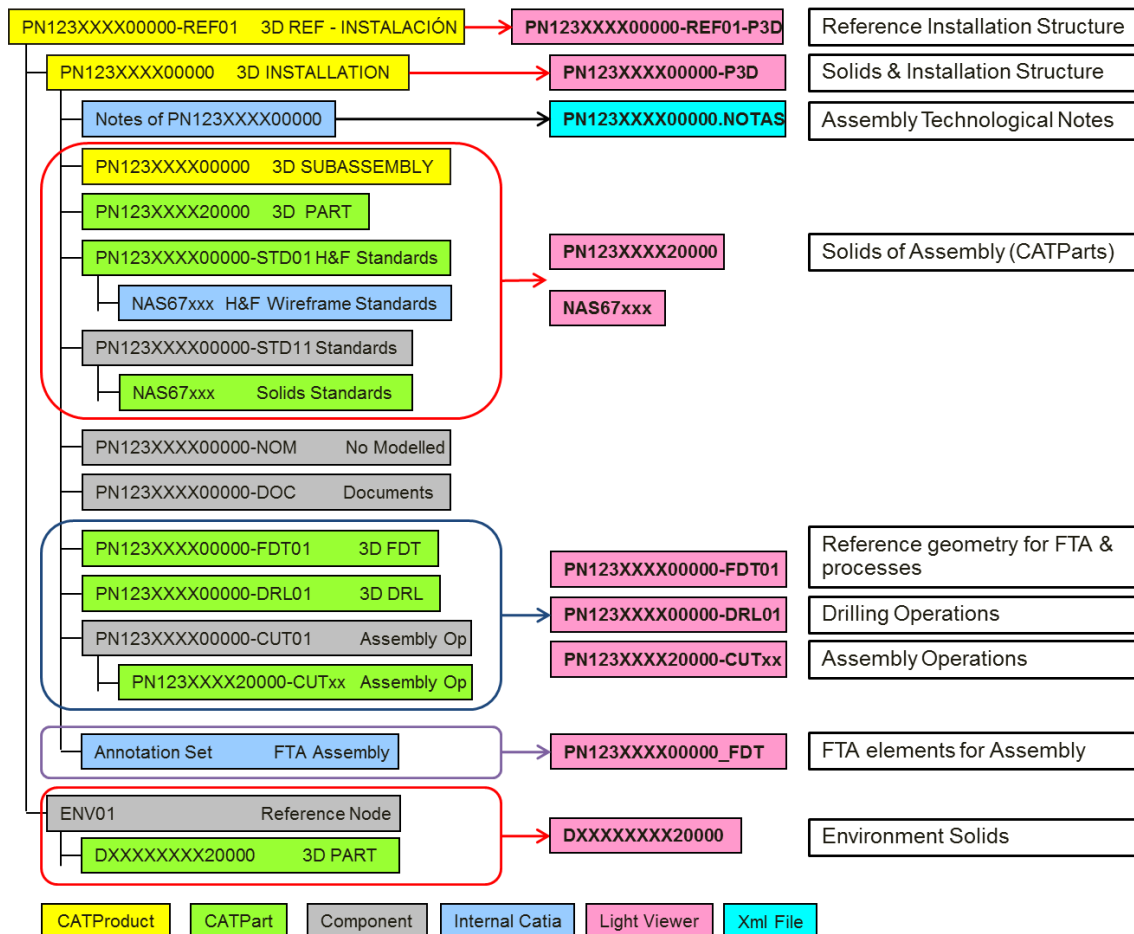




ASSY

Data of Element/Component – Installation assembly

Installation: Represent the allocation of assemblies and so, it is not storable



ii Many Thanks for your attention !!

